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Spatio-temporal network modeling and analysis of global strong earthquakes ($M_w \geq 6.0$)

Gang Liu^{1,2}, Qinjin Fan³, Weile Li^{1,*}, Gianvito Scaringi⁴, Yujie Long², Jing He², Zheng Li⁵

¹ State Key Laboratory of Geohazard Prevention and Geoenvironment Protection, Chengdu University of Technology, Chengdu 610059, China.

² College of Earth Sciences, Chengdu University of Technology, Chengdu 610059, China.

³ Department of Geography, University of Georgia, Athens, GA 30602, USA.

⁴ Institute of Hydrogeology, Engineering Geology and Applied Geophysics, Faculty of Science, Charles University, Prague 128 43, Czech Republic.

⁵ Sichuan Provincial Land and Space Planning Research Institute, Chengdu 610084, China.

 G.L., 0000-0002-8856-8746; W.L., 0000-0002-3741-8801; G.S., 0000-0003-3505-7456

* Correspondence: Weile Li, whywl01@163.com

Abstract: We employ a spatio-temporal network modeling approach to identify possible relations among strong earthquakes and spatial regions worldwide. A global strong earthquake dataset containing 7,736 events ($M_w \geq 6.0$) from 1964 to 2018 is used. Statistical results identify power law relationships and heavy tail phenomena in the spatial patterns of strong earthquakes. The interactions between different regions follow the same law, with a few regions that may be hit by successive strong earthquakes with high probability. Besides, we find that the interconnections among regions is mainly related to the succession of events in time, whereas it is extremely inhomogeneous in space. This study provides a research prototype for the spatio-temporal analysis of global strong earthquakes, laying a foundation for obtaining insights into the network modeling approach of global strong earthquakes.

Our planet was affected by more than 11,000 strong earthquakes (moment magnitude $M_w \geq 6.0$) during the period 1900–2018 (USGS, 2018). Over 2,900 earthquakes occurred in the 21st century, which caused more than 800,000 direct fatalities (USGS, 2018). With the global population rapidly increasing, the risk posed by earthquakes and earthquake-induced chains of geological hazards (Fan et al., 2018) to human lives, properties, and infrastructures is expected to increase, especially in fast-developing countries (e.g., Huppert and Spark, 2006), even though fatality rates are decreasing thanks to an increasing preparedness to natural catastrophes (Bilham, 2004).

Predictable patterns of seismicity can be identified at some scales (Bak et al., 2002; Corral, 2003; Saichev and Sornette, 2007; Marekova, 2012). On the other hand, our incapability of predicting where and when the next strong earthquake will strike – in such a way that effective early warning can be enforced – is perceived as an issue among the population and the policy-makers (Davis et al., 2010), and is a source of frustration among scientists (Joffe et al., 2017). Within these uncertainties, it can be argued that the best strategy to cope with the earthquake risk is to either focus only on improving the preparedness, or to jointly optimize prediction and

preparedness to optimize the cost/benefit ratio (Davis et al., 2010). A better understanding of earthquake patterns remains undoubtedly necessary, in any case.

The pioneering works of Gutenberg and Richter (1944) and Omori and Coll (1894) shed the first lights on the temporal patterns of earthquakes and their aftershocks in a given area. Works such as that of Burridge and Knopoff (1967) revealed the role of friction and viscosity in the earthquake mechanism and seismic sequences. As summarized by Chakraborty et al. (2015), subsequent studies began looking at earthquake dynamics as scale invariant processes. Correlations among mainshock and aftershocks, and the distribution of recurrence times distributions have been investigated (e.g., Bak et al., 2002; Christensen et al., 2002; Corral, 2003, 2004). It has been also argued that the locations of earthquake epicenters can form fractal sets (Okubo and Aki, 1987; Shapiro and Faizullin, 1992; Kagan, 1994), i.e. scale-invariant distributions. Even though the probability of occurrence of two strong earthquakes in the same area in a narrow time window may seem very low, the interval between any two successive events is highly variable, and deterministic forecasting has not yet become a reality (Chakraborty et al., 2015).

Large earthquakes can trigger other earthquakes elsewhere. Researchers observed that large aftershocks tend to occur close to the mainshock, whereas microearthquakes can be triggered by the travelling seismic waves at significant distances from the mainshock (Hill et al., 1993; Brodsky et al., 2000; Kilb et al., 2000; Gomberg et al., 2001; Parsons, 2002; West et al., 2005; Parsons & Velasco, 2011). However, Brodsky et al. (2000) speculated that dynamic triggering on a regional scale can result in countrywide episodes of increased seismicity. Parsons and Velasco (2011) found that $M_w > 5$ earthquakes can be triggered by any $M_w > 7$ mainshock. Pollitz et al. (2012) suggested a Love wave-related mechanism responsible for remotely (>1500 km) triggering aftershocks ($5.5 \leq M_w \leq 7$) after the 2012 M_w 8.6 east Indian Ocean earthquake. The authors also speculated that the unusually high number of these aftershocks could be related to the strike-slip nature of the mainshock or the relatively low global earthquake rate in the period that preceded it. More broadly, Sammis and Smith (2013) defined a system of non-linear coupled relaxation oscillators to demonstrate how the communication between large earthquakes on a global scale can align their seismic cycles to produce a worldwide clustering of large events (Sammis and Smith 2013). On the other hand, researches indicate that the temporal distribution of global strong earthquakes is well described by a random process, and apparent clustering is only due to random variability (Michael, 2011).

In recent years, the spatio-temporal complexity of seismic data has become an object of study of the Complex Network Theory (CNT) (Baiesi and Paczuski, 2004; Abe and Suzuki, 2004, 2007; Abe et al., 2011; Chakraborty et al., 2015; Liu et al., 2018). It is believed that most real systems can be abstracted as a network, so that their topological and functional properties can be studied (Albert and Barabási, 2002; Newman, 2003; Arenas et al., 2008; Barthélemy, 2011). Networks associated with earthquake patterns show *scale-free* and *small-world* properties (Baiesi and Paczuski, 2004; Abe and Suzuki, 2004, 2007; Abe et al., 2011; Chakraborty et al., 2015; Liu et al., 2018).

A CNT approach has been used, for instance, by Jiménez et al. (2009) to discriminate an aftershock cluster from other clusters of regional seismicity in the Pyrenees – associated with different seismic sources – using network properties and fractal dimension. Again Jiménez (2013) applied a CNT strategy using Shannon's (1948) concept of mutual information to explore the processes producing certain seismic patterns in California. Other examples of regional seismicity investigated through complex networks are given in Daskalaki et al. (2014,

2016) for case studies in Greece and Italy, in León et al. (2018) for a case study in Colombia, and in Pastén et al. (2018a, 2018b) for several data sets in Chile. The authors also argued that some parameters of the networks may be site-specific rather than universal, notwithstanding the general *scale-free* properties of the networks.

Abe and Suzuki (2004) proposed a method for earthquake network modeling in which a region is divided into regular cells of a given size (e.g., 10×10 km) and each cell is abstracted as a vertex. A link connecting two vertices is created when two successive events occur in the respective cells. Then, a time-space network describing seismic data is constructed. This network is helpful to study the strength of interactions among different cells. By accounting for lithological information on the subsurface, Liu et al. (2018) proposed a different approach for constructing earthquake networks in which a rock mass, rather than a grid cell, is regarded as a vertex, and a link connecting two vertices is generated when two successive events occur in distinct rock masses. This approach can be helpful to reveal geo-lithological interactions among lithotypes at regional or local scales.

In this work, we analyze the spatio-temporal dynamics of global strong earthquakes using a CNT approach. To this aim, we construct a strong-earthquake network as a weighted network using spherical triangles to identify the network cells. Given the apparently random spatio-temporal variability in the occurrence of strong earthquakes, we investigate whether patterns exist, that can be identified through a CNT approach at global scale.

Definition of symbols

Key mathematical symbols used in the paper are defined in **Table 1**.

Methodology

The seismic catalog of $M_w \geq 6.0$ earthquakes for the CNT analysis must be a complete one so that the true succession of events can be identified. Previous research demonstrated that the available catalogues were incomplete at $M_w \geq 6.0$ until 1964, when the World-Wide Standardized Seismograph Network became largely operative (Michael, 2014). Therefore, we only use the USGS catalog of $M_w \geq 6.0$ events from 1964 onwards. Until the end of 2018, 7,736 strong earthquakes ($M_w \geq 6.0$) were recorded globally (USGS, 2018). The spatial distribution of these earthquakes is shown in **Figure 1**. Unsurprisingly, a large number of earthquakes occurred around the plate boundaries, frequently in coastal zones. An approach to investigate possible patterns is to consider the whole planet as one complex system, with the strong earthquakes being regarded as the result of interactions between components of the system.

To identify possible relationships between earthquakes, Abe and Suzuki (2004) proposed a method for creating earthquake networks, in which an earthquake-prone region is discretized into a geographical square grid. However, this approach is inconvenient to study earthquake patterns at global scale due to Earth's curvature. Therefore, we use an icosahedron-based subdivision (Teanby, 2006) using spherical triangles.

Let R be the radius of the Earth, d the tessellation level, and D the spatial scale or resolution (which equals to the average length of the sides of triangles), the tessellation method can be described as follows:

- (1) Firstly, an icosahedron is used to represent the globe and the tessellation level d is

defined as 0.

- (2) For the next tessellation level $d = d + 1$, for each triangle of the regular polyhedron below the last level, the midpoints of three edges are computed and then projective points of three midpoints on the sphere are obtained, imposing the gnomonic projection rule.
- (3) The three projective points are connected to each other and also each point is connected to corresponding two ending points, each triangle in last level is thus subdivided into 4 smaller triangles, and a new, higher-resolution triangle grid system is generated.
- (4) Steps (2) and (3) are repeated recursively, until the global tessellation grid at the required level is obtained. For simplicity, we call these spherical triangles *triangle cells* or simply *cells*.

When a global tessellation grid at the desired level is obtained, the relationship between the cells and the earthquake events can be determined. Each earthquake is attributed to a single cell, while the cells may be associated to zero, one, or multiple earthquakes. In this way, we are able to see the spatial distribution of earthquakes across the cells. Then, to understand whether interactions between different cells exist, we investigate the temporal pattern of the earthquakes. The occurrence of successive events can be tracked as a spatio-temporal earthquake network system, in a way that each earthquake has a definite position and time. Assuming that N strong earthquakes occurred within a time interval, and the i_{th} earthquake occurred at time t_i , the basic principles to construct the network are as follows:

- (5) Each cell, in which at least one earthquake occurred, is identified as a vertex (also termed *node*).
- (6) A directional link is generated between two vertices if two successive events occurred at these two vertices; the direction of the link is from the vertex of the i_{th} earthquake (the *source*) to the vertex of the $(i + 1)_{\text{th}}$ earthquake (the *destination*). A loop is generated, i.e. a link from a cell pointing to itself, if two successive events occurred in the same cell.
- (7) The weight assigned to a vertex is defined as the total number of strong earthquakes that occurred within the corresponding cell, and is represented by v . The weight reflects the “activeness” of earthquakes within this region.
- (8) The weight associated with a specific link is defined as the number of successive earthquakes between its two end vertices, and is represented by w . The link weight can be considered as the “interaction strength” (or correlation degree) between two regions. A large w indicates that successive events between two regions occurred with comparatively high probability.

Figures 2a–c exemplify the modeling principle of the spatio-temporal earthquake network: in (a), a global tessellation grid with 320 triangle cells is shown; in (b) the locations of the 7,736 strong earthquakes in the dataset is overlaid on a tessellation grid with 5,120 cells; and in (c) the modeling process of the spatio-temporal earthquake network is shown, using 19 successive events. In this model, a coordinate is associated to each vertex, which corresponds to the center of the cell. Besides, each vertex has a connectivity (or degree) which is defined as the number of other vertices directly connected to it, and is represented by k . **Figure 2d** presents the 2-dimensional map of the earthquake network overlaid on a global remote sensing image based on the tessellation grid in **Figure 2b**. The vertices on Solomon Islands, Japan, Taiwan, North Maluku, and Northeastern Afghanistan are of larger size and darker color, which indicates that more earthquakes occurred in these regions; therefore, there are also more connections between

these and other regions. It can also be seen that most of $M_w \geq 6.0$ earthquakes occurred around the continent margin of the Pacific Ocean, and many interconnections exist among vertices generated in this region (**Figure 2d**).

Results and discussions

We constructed global networks at various spatial resolutions. **Figure 3** indicates that the distributions of vertices and link weights both present power-law scaling, and the exponents of both distributions tend to increase with D decreasing. We can use $P(v) \propto v^{-\gamma}$ and $P(w) \propto w^{-\lambda}$ to describe the distributions of vertex weights and link weights, respectively. It is clear that if the area of a cell becomes smaller, it will tend to contain less events. Therefore, the ratio of vertices with small v and the ratio of links with small w will both increase; conversely, the ratio of cells with large v and w will decrease. **Figure 3a** indicates that the distribution of strong earthquakes is extremely inhomogeneous in space, so that there are a few regions that are hit by a large number of events. **Figure 3b** suggests that the strength of the interconnections (or interactions) among regions is unevenly distributed.

Figure 3 illustrates the relationships between temporally successive events across the globe. In order to further study the spatio-temporal features of these events, earthquake networks for different regions are rebuilt and corresponding indices are computed. For instance, **Figure 4** shows the distributions of vertex weights and link weights for Southeast Asia and South America. In this case, only the events occurred within the region are used to construct the spatio-temporal earthquake network. Again, the distributions of vertex weights and link weights both obey power law distributions, which are similar to those at global level. It is worth noting that, under such distributions, vertex or link weights may achieve large values with a non-negligible probability. From **Figure 3** and **Figure 4**, it can be observed that both the spatial distribution of the events and the interactions between the cells are strongly uneven. In fact, while many cells contain few earthquakes and the relationships between most of the cells are small, some cells stand out for the large number of events within them as well as for the strong interactions with some other cells.

The weight of vertices (i.e. regions) obviously depends on the spatial distribution of earthquakes: the more earthquakes occur within the region, the greater will be the weight associated to it. Conversely, we consider that the weight of links should be mainly related to the temporal distribution of earthquakes, since a large number of successive events influences the distribution pattern of link weights (i.e. the interconnections among regions). Therefore, we believe that the interconnections among regions resulting from this paper are not an artifact (i.e. due mainly to the temporal succession of events and the presence of areas with a large number of events that are, in fact, not related to each other in any way).

In order to demonstrate this perspective, it is necessary to apply exactly the same methodology to events that are not temporally correlated at all. With this in mind, the simplest way to do this is to shuffle the earthquake catalog in time (i.e. changing only the order of events, but not the times of occurrence), thus destroying the potential temporal correlations. The shuffling process is detailed as follows:

- (1) The actual events are numbered (i.e. from 1 to 7,736) in chronological order, so that each number corresponds to a unique time;

- (2) All the events' numbers are regenerated randomly with non-duplicated numbers within the 1–7,736 range;
- (3) The times of occurrence for the renumbered events are assigned based on the corresponding relation between number and time established in step (1);
- (4) The spatial information of all the events remains unchanged during the whole shuffling process. According to the process, we see that although the events are shuffled, the temporal density and the spatial aspects of the patterns would be preserved. Then, we can compare the results to those obtained for the original data.

With this in mind, we construct the earthquake networks at different spatial scales based on the shuffled data. Because link weights reflect the strength of interconnections among regions, we study the relation between link weight w and vertex weight v of two end vertices connected to a link. **Figure 5** indicates that the results based on the original successive events are obviously different from those based on the shuffled events. **Figures 5a–c** illustrate that when the source weight and destination weight are both large, the link weight can be large too. Most significantly, we observe that the link weight is large if the weights of the source and destination are approximately equal. If one end vertex's weight is large, but the other one's is small, the corresponding link weight might be very small. Therefore, we see that the link weights are very large along the diagonal line direction of both source weight and destination weight increasing.

Figures 5d–f show that for the results based on the shuffled data, the maximum link weight is much less than that of the actual data. For example, when $D = 483.6$ km, the maximum link weight of the actual data is $w = 35$, which is 5 times the maximum link weight of the shuffled data ($w = 7$). Besides, the relation of link weight and two end vertices' weights for the shuffled data is obviously different from that for the actual data. **Figure 5a** indicates that, for the actual data, the link weight is great when the two end vertices' weights are both great and the difference between them is small. Thus, most of greater values of link weight occur along the direction in which two end vertices weights increase synchronously. **Figure 5d** illustrates that, for the shuffled data, the link weight approximately goes down annularly along the direction from the maximum weight of two end vertices to the minimum weight. It can be considered that, for the shuffled data, the link weight is proportional to the product of two end vertices' weights. However, for the actual data, the link weight is not only proportional to the product of two end vertices' weights, but is also inversely proportional to the difference between them.

We find that the results from the shuffled data are significantly different from the results based on the actual data. The relations or interconnections among regions can be thus clearly revealed using the spatio-temporal earthquake network based on the actual successive events. Besides, it also demonstrated that the connection strength among regions is mainly related to the temporal succession of events, and to the presence of areas with a large number of events.

To further clarify the statistical characteristics of the seismic dataset, we compare the actual data with a null hypothesis, defined as follows:

- (1) For a given tessellation scale D , the non-empty cells in the seismic dataset are chosen for modeling, where these cells form a collection represented with Θ ;
- (2) N events (7,736) are generate orderly by randomly choosing the locations (i.e. cells)

- (3) A spatio-temporal earthquake network is built based on these randomly created events according to the proposed approach.

For a given tessellation scale D , this null hypothesis was simulated 100 times and the average result was chosen for investigation. **Figure 6** shows the distributions of vertex weights and link weights for the null hypothesis. The result indicates that $P(v)$ follows a Gaussian distribution, which can be described as $P(v) = (0.133 \pm 0.004) \cdot \exp^{-\left(\frac{v-(8.990 \pm 0.087)}{4.251 \pm 0.123}\right)^2}$, while $P(w)$ satisfies an exponential distribution so that $P(w) = (79.400 \pm 0.710) \cdot \exp^{(-4.387 \pm 0.009) \cdot w}$.

Conversely, for the actual seismic dataset, vertex weights and link weights were both found to follow power law distributions. The probability of attaining large values for a power law distribution is obviously larger than that of a Gaussian distribution, and it can be inferred that the distributions of vertex weights and link weights for the actual spatio-temporal distribution of strong earthquake events are different from those of the null hypothesis. In other words, the actual succession of events cannot be randomly generated.

In order to further demonstrate that the interconnections among regions are mainly related to the succession of events in time, we respectively divide the original data and shuffled data into three subsets with approximately the same number of events. Based on this, we construct the earthquake network of each subset and study the distribution of link weights in these subsets. The results, as shown in **Figures 7a–d** (with tessellation scale $D = 120.8$ km), indicate that, for the original succession of events, the link weights of the subsets all present remarkable power-law distribution. The maximum link weight of the spatio-temporal earthquake networks of the subsets is 10, while the maximum link weight of the original spatio-temporal earthquake network is 18. **Figures 7e–h** show that for the shuffled data, the value of link weight is very poor. For example, the earthquake network in **Figure 7e** contains 2,568 links, but the weights of 2,558 interconnections among regions are equal to 1 and only 10 links' weights are equal to 2. The maximum link weight of the earthquake network constructed by considering all of the 7,736 shuffled events is equal to 3, only one link attains the maximum weight, as seen in **Figure 7h**, which means that the interconnections among regions are all very poor.

Significantly, **Figure 7h** indicates that, for the shuffled data, the distribution of link weights shows an exponential distribution rather than a power law. Although the shuffled events still have the spatial information of the actual events, they do not preserve the temporal relationship between actual events, meaning that the shuffled events are not temporally correlated. Since the links are generated in chronological order, the spatio-temporal characteristics of the actual events are lost upon shuffling. Therefore (see **Figure 5** and **Figure 7**), it can be concluded that the spatio-temporal characteristics of global strong earthquakes are controlled by the actual succession of events in time, and their uneven distribution in space. Going back to the assumption of the Earth as a complex system, these results suggest that the interactions or activities inside the Earth's crust are not totally independent in space and time, and the relations identified between successive events may reflect the spatial-temporal laws of crustal movements. Moreover, by constructing the global earthquake networks over different time periods, we find that the resulting earthquake networks are all described by similar spatial-temporal laws.

For further investigation, we generated earthquake networks for different periods and tessellation scales. We then calculated key topological measures for each network. **Table 2**

shows the values of six topological indices for each network, and **Figure 8** plots them as a function of time.

The average connectivity of a network is an important structural feature, usually represented by $\langle k \rangle$. As shown in **Figure 8a**, $\langle k \rangle$ presents an approximately linear increasing trend with time, which indicates that newly created links are more possibly connected to the vertices with high degrees (as expected).

The power law exponent is another significant characteristic: we computed the exponents γ and λ respectively for the vertex and link weight distributions. **Figures 8b** and **8c** show that both the exponents decrease with time. However, it is possible that they are actually tending to constant values, which could be revealed if a sufficiently large time window is used. This would mean that the distribution of large global earthquakes tends to stabilize, and extreme changes should not occur. This conclusion is consistent with Michael's (2011) viewpoint that the likelihood of future events is not increasing.

The clustering coefficient is a measure of the degree to which nodes in a graph tend to cluster together, which is the number of actual connections across the neighbors of a particular node, as a percentage of possible connections. The average clustering coefficient of a network reflects the local "tightness" of the network, represented by $\langle C \rangle$. **Figure 8d** indicates that the average clustering coefficient of the global strong earthquake network increases over time. This phenomenon suggests that the local structure of the earthquake network becomes tighter with time, meaning (as expected) that more earthquakes occur in regions already hit by earthquakes in the considered time window.

The modularity measures the degree of correlation between the probability of having an edge joining two nodes and the fact that the nodes belong to the same community (Albert and Barabási 2002), represented by Q . **Figure 8e** indicates that the modularity Q of the global strong earthquake decreases with time. It means that the community structural strength of the earthquake network is gradually weakening.

In **Figure 8f**, the average path length $\langle L \rangle$ of the network shows a decreasing trend with time and presents a remarkable *small-world* effect. The experiments over time periods all indicate this phenomenon. The *small-world* effect refers to the observation that most pairs of vertices in the network are connected by a short path through the network. This phenomenon again indicates that new earthquakes tend to occur in regions already hit by earthquakes in the past rather than in new regions. Taking the tessellation scale $D = 241.9$ km as an example, statistical results show that about 9.1% successive events occurred in the same cell and many other successive events occurred in adjacent cells. In **Figures 8b**, **8c**, and **8f** we can see that the patterns of $P(v) \propto v^{-\gamma}$ and $P(w) \propto w^{-\lambda}$ tend to stabilize, suggesting on the contrary that the global spatio-temporal strong earthquake network is not well defined in the short term. Instead, its evolution may be measured or has definite spatio-temporal features in long term observations. If additional factors are taken into account in the modelling (such as the earthquake interoccurrence time, surface distance, stress, and focal depth), we speculate that more seismic dynamics could be revealed. For instance, it could be explored whether the spatio-temporal earthquake network modelling can support the study of remote triggering of large earthquakes.

Figure 8 illustrates some regularities in the evolution of the earthquake network through time (i.e. as the number of data points increases). It can be interesting to verify whether some of the observed trends can also be found in earthquake networks constructed for independent periods

with approximately the same number of events. To this aim, we again divide the seismic dataset into three independent subsets, containing 2,578, 2,579, and 2,579 events.

The topological measures of the spatio-temporal earthquake networks for these three subsets under different tessellation scales D are shown in **Table 3**. For a given D , the networks corresponding to the subsets are approximately equivalent in size and there are no statistically significant differences in the number of nodes and edges between the networks. The results indicate that, for a given D , the topological measures of the three networks are to some extent different. This suggests that it may be difficult to infer the measures of a network for an independent time period based on the measures of a known network. On the other hand, as demonstrated earlier, the topological measures present significant trends over cumulated time periods.

Conclusions

Using the Complex Network Theory, we defined a modeling approach for constructing a global strong earthquake network. We divided the Earth's surface into cells based on spherical triangles and used a global strong earthquake dataset ($M_w \geq 6$) from 1964 to 2018 to construct the network and analyze its spatio-temporal characteristics. The relations between different regions that can be obtained present power law distributions. Strong interactions between some regions are apparent, and the number of these regions is non-negligible. Besides, the spatial distribution of strong earthquakes is also uneven and shows power distributions and heavy tail phenomena. We analyzed several topological measures to identify possible patterns. We show that the spatio-temporal features of strong earthquakes and the interconnections among regions mainly stem from the succession of events in time and the uneven distribution of events in space. Our work constitutes a novel approach for the modeling and analysis of the spatio-temporal complexity of global strong earthquakes. The method can be used to mine the spatial distribution and trends of strong earthquakes. Besides, this work provides some references for the study of remote triggering of large earthquakes. However, further study is required to identify possible references or trends that may be usable for predictions.

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Figure 1. Spatial distribution of global strong earthquakes ($M_w \geq 6.0$) (Map data: Google Earth, © 2020 NASA, TerraMetrics). The epicenters are marked by yellow dots. The dataset was obtained from the United States Geological Survey (USGS, 2018) and contains 7,736 records from January 1st, 1964 to December 31st, 2018.

Figure 2. Schematic description of the spatio-temporal earthquake network. The Earth surface can be subdivided into equilateral spherical triangles as shown in (a), which contains 320 triangles. After successive subdivisions, finer grids can be obtained, composed of more triangles. The number of triangles after d subdivisions equals to $20 \cdot 4^d$. For any subdivision level, the tessellation scale D can be computed, so that it is equal to the average length of the sides of triangles. For example, if the radius of the Earth is set to 6,371.0087714 km (the average radius under WGS-84 coordinate system) and $d = 4, 5, 6$, the total number of triangles is respectively equal to 5,120, 20,480, and 81,920; the corresponding spatial scale D is 483.6, 241.9, and 120.8 km. In (b), a grid with 5,120 triangles is shown, together with the locations of the 7,736 strong earthquakes in the database. In (c), the modeling principle of the spatio-temporal earthquake network is shown. Links are generated to connect two regions where two successive events occurred. In (c), 19 earthquakes occurred in 12 regions, which are numbered from 1 to 12. According to their temporal succession, the sequence is described as $1 \rightarrow 9 \rightarrow 3 \rightarrow 8 \rightarrow 7 \rightarrow 5 \rightarrow 9 \rightarrow 11 \rightarrow 12 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 8 \rightarrow 8 \rightarrow 1 \rightarrow 10 \rightarrow 9 \rightarrow 2 \rightarrow 3$. Additionally, 18 links are defined $\{<1, 9>, <9, 3>, <3, 8>, <8, 7>, <7, 5>, <5, 9>, <9, 11>, <11, 12>, <12, 4>, <4, 5>, <5, 6>, <6, 8>, <8, 8>, <8, 1>, <1, 10>, <10, 9>, <9, 2>, <2, 3>\}$. In (d), a 2-dimensional visualization of the spatio-temporal earthquake network is shown, in which the spatial scale of triangular subdivision is 483.6 km (i.e., the total number of triangles is 5,120; Map data: Google Earth, © 2020 NASA, TerraMetrics).

Figure 3. Distributions of vertex weights and link weights for different spatial scales. $D = 483.6, 241.9, 120.8$ km respectively represent the average length of the sides of triangle cells for tessellation levels $d = 4, 5, 6$, for which the total number of cells is respectively equal to 5,120, 20,480, and 81,920.

Figure 4. Distributions of vertex weights (a, c) and link weights (b, d) with $D = 120.8$ km for Southeast Asia (a, b) and South America (c, d). Only the events occurred within the region are considered for constructing the spatio-temporal earthquake network.

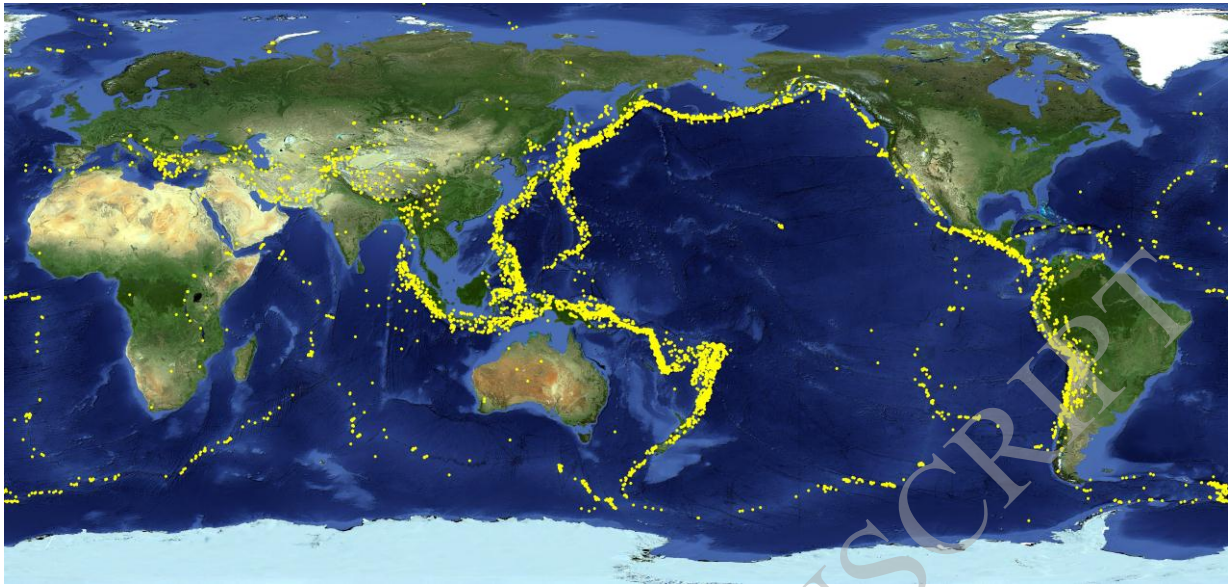
Figure 5. Contour map of the scatter plot for correlation between link weight w and vertex weight v . Each link has two end vertices (i.e. source and destination), so this plot reflects the change of link weight with both the weights of two end vertices. The colors from green to red indicate the change of link weight w from small to large. (a–c) Results for spatial scales $D = 483.6, 241.9, 120.8$ km, respectively, based on the original successive events data. (d–f) Results for the same spatial scales based on the shuffled data.

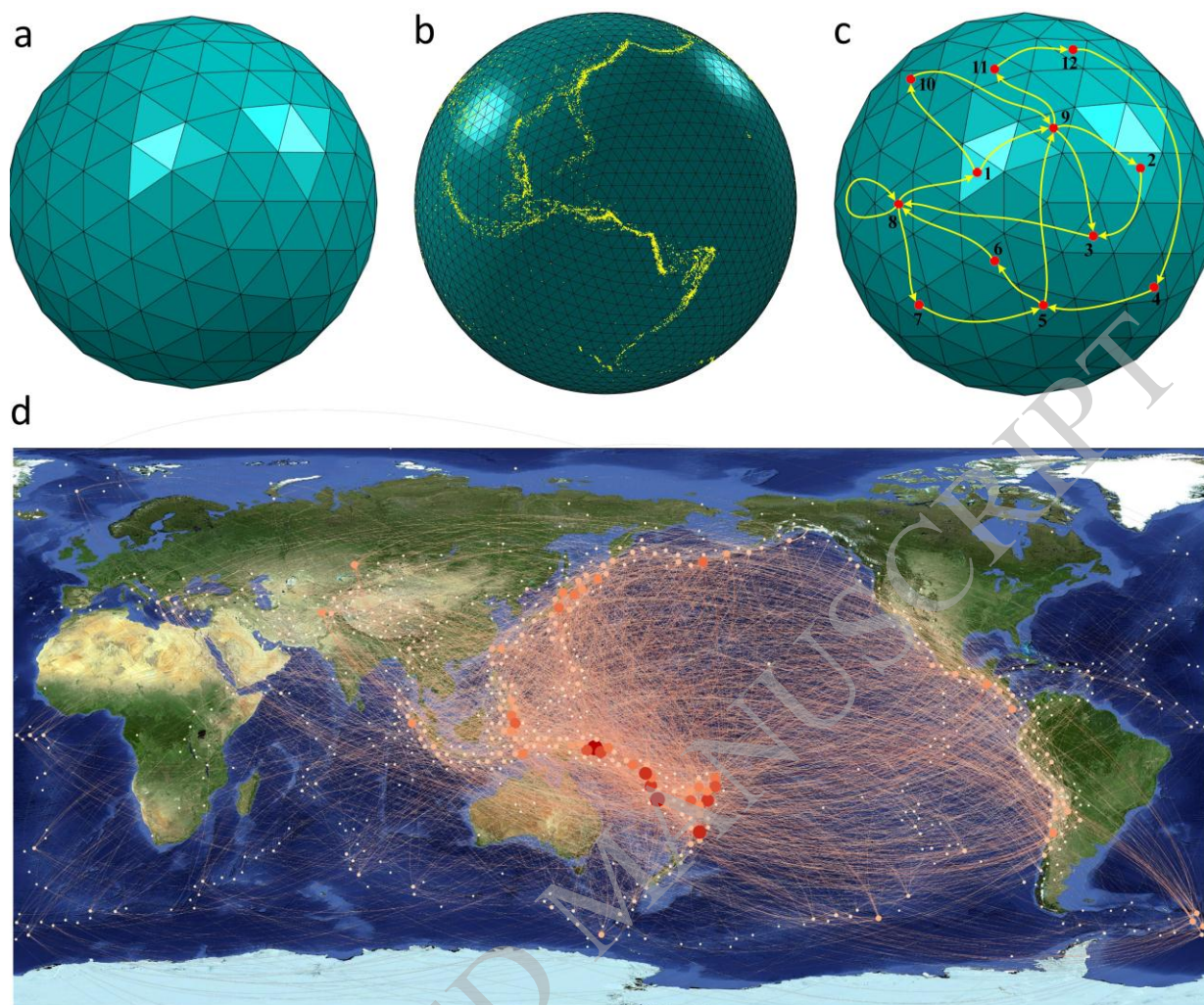
Figure 6. Distributions of vertex weights and link weights for the null hypothesis when the tessellation scale $D = 483.6$ km. In (a), the distribution of vertex weights is shown, which can be well represented by a Gaussian function, $P(v) = (0.133 \pm 0.004) \cdot \exp^{-\left(\frac{v - (8.990 \pm 0.087)}{4.251 \pm 0.123}\right)^2}$. In (b), the distribution of link weights is shown, which can be well represented by an exponential function, $P(w) = (79.400 \pm 0.710) \cdot \exp^{(-4.387 \pm 0.009) \cdot w}$.

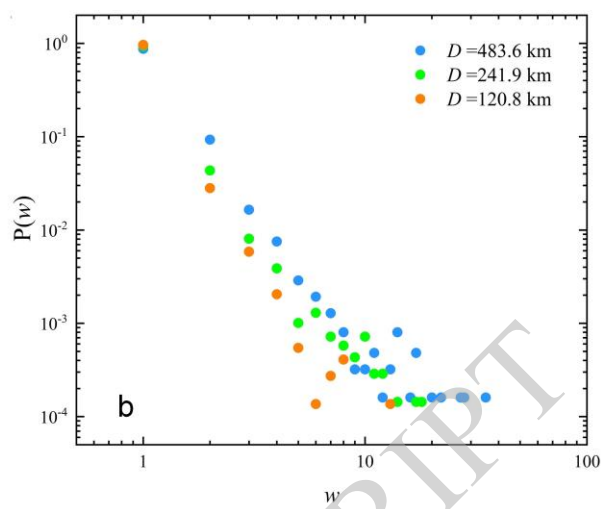
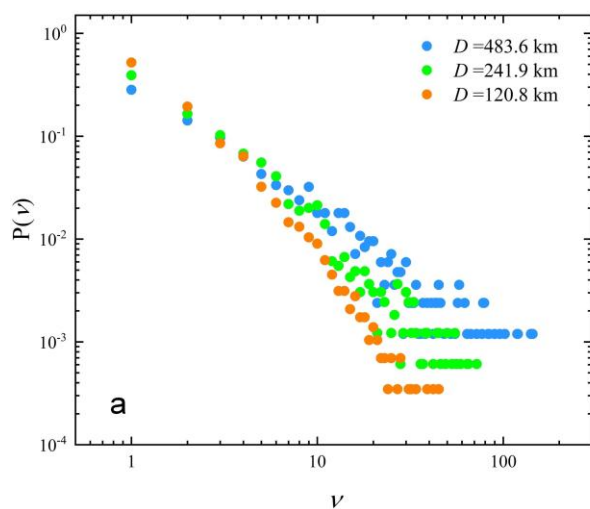
Figure 7. Distributions of link weights w for the original and the shuffled data for $D = 120.8$ km. The original successive events are approximately equally divided into three subsets based on time: subset 1 from 1 to 2,579, subset 2 from 2,580 to 5,158, and subset 3 from 5,159 to 7,736. For the shuffled data, the same division method is chosen. (a–c) are the distributions of link weights for the subsets of the original data, while (d) refers to the entire dataset (7,736 earthquakes). It is easy to see that the events in the subsets divided based on the original data are successive in time, the succession between any two

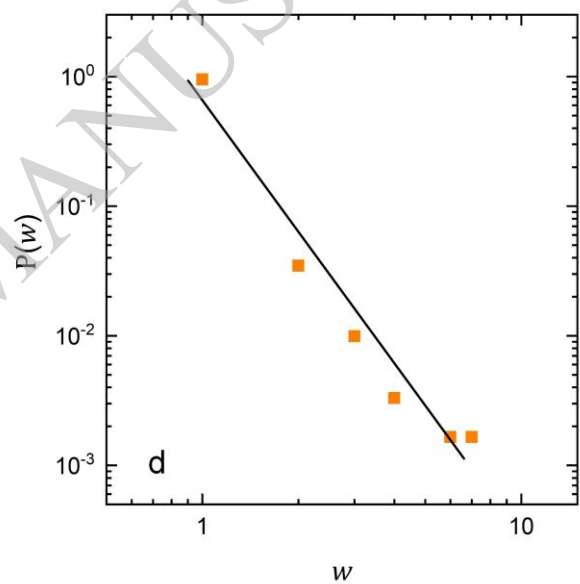
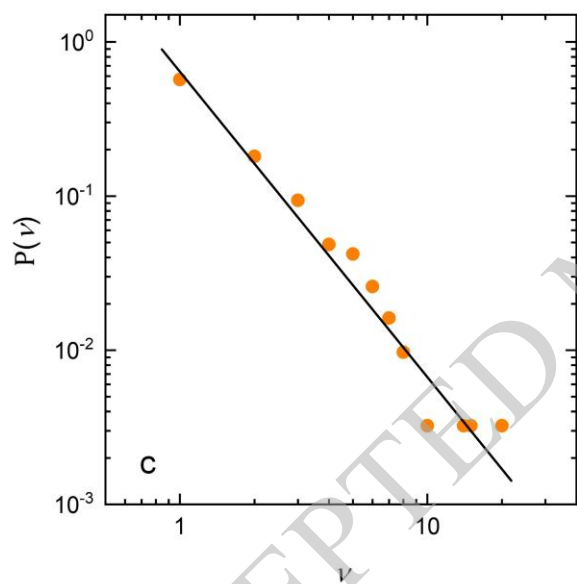
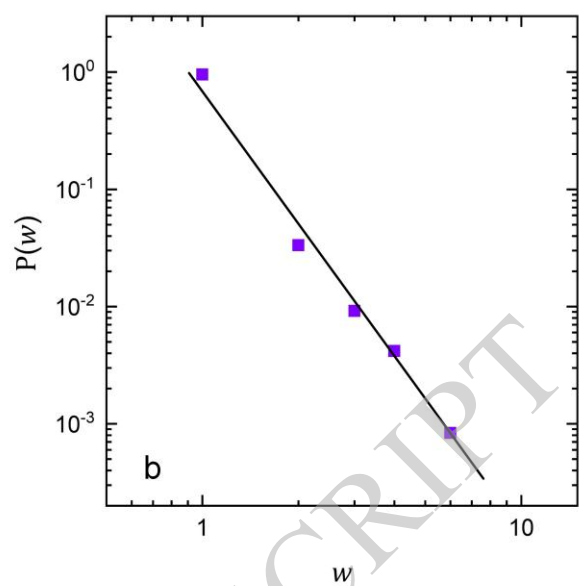
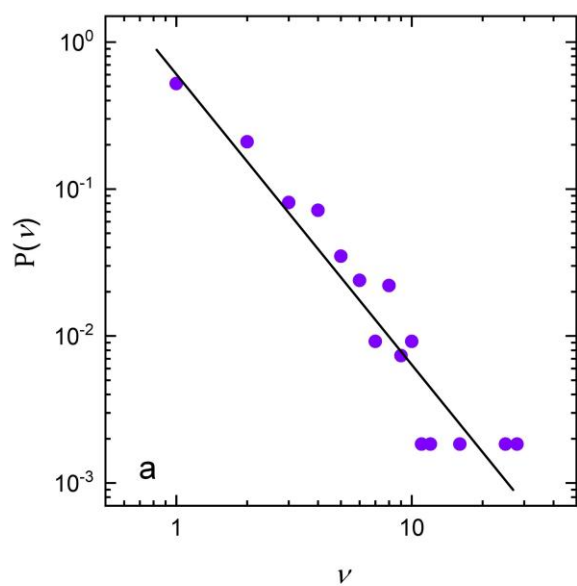
events in the subsets is the same as that in the original data. In (d), the distribution of link weights shows a power law, $P(w) = 0.963 \cdot w^{-5.071}$. (e–g) are the distributions of link weights for the three subsets of the shuffled data, while (h) refers to the entire shuffled dataset. In (h), the distribution of link weights can be well represented by an exponential function, $P(w) = 190.300 \cdot \exp^{-5.254 \cdot w}$.

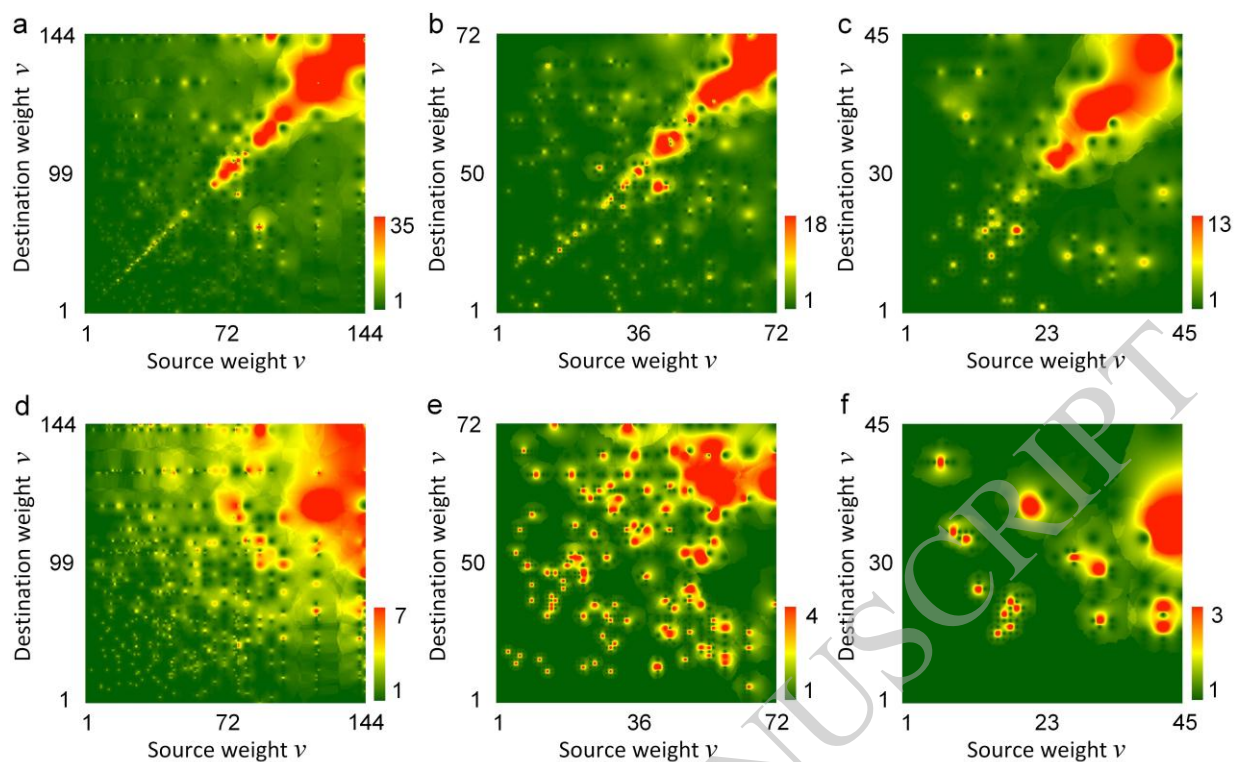
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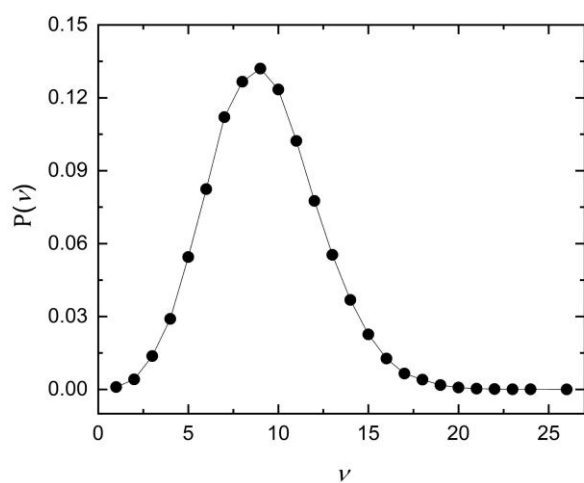




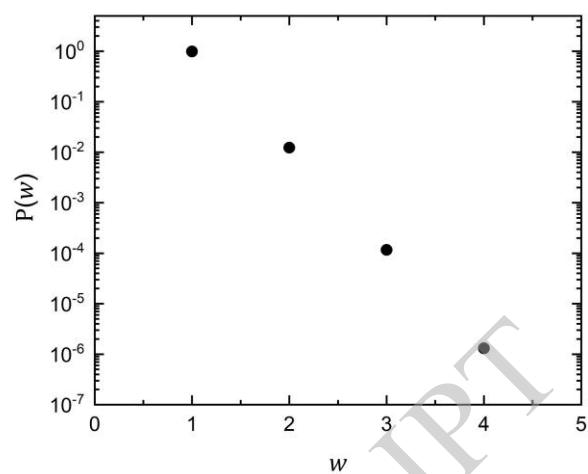




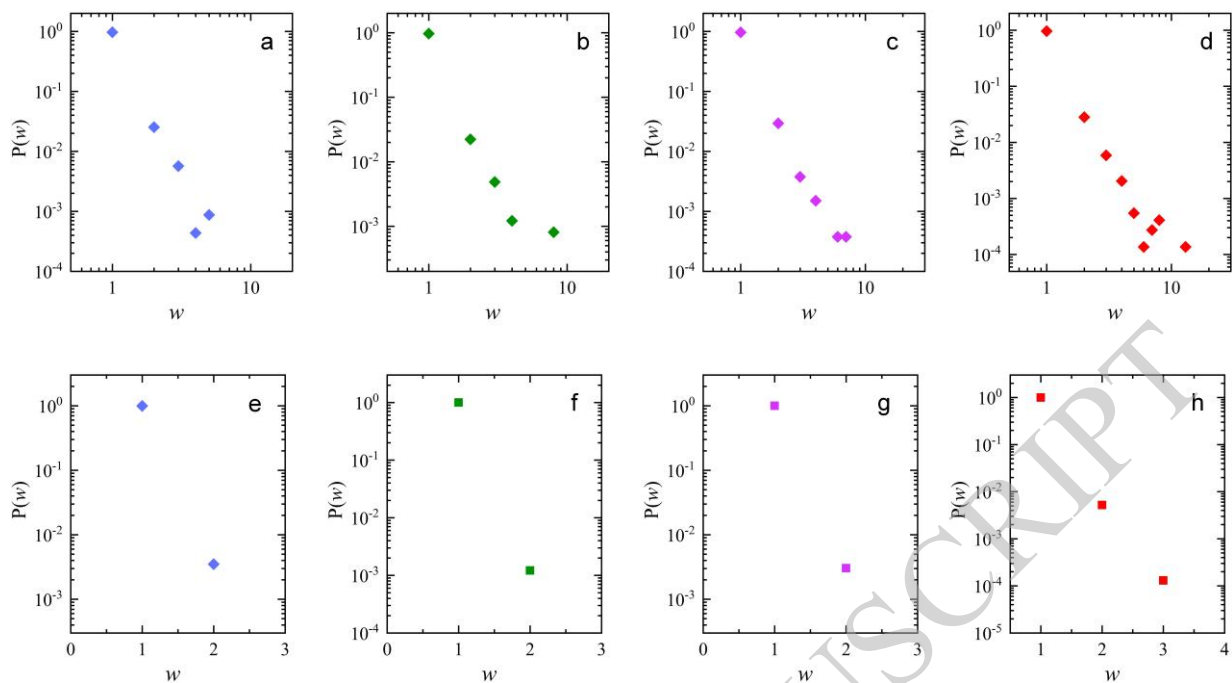




(a) $P(v)$ vs v



(b) $P(w)$ vs w



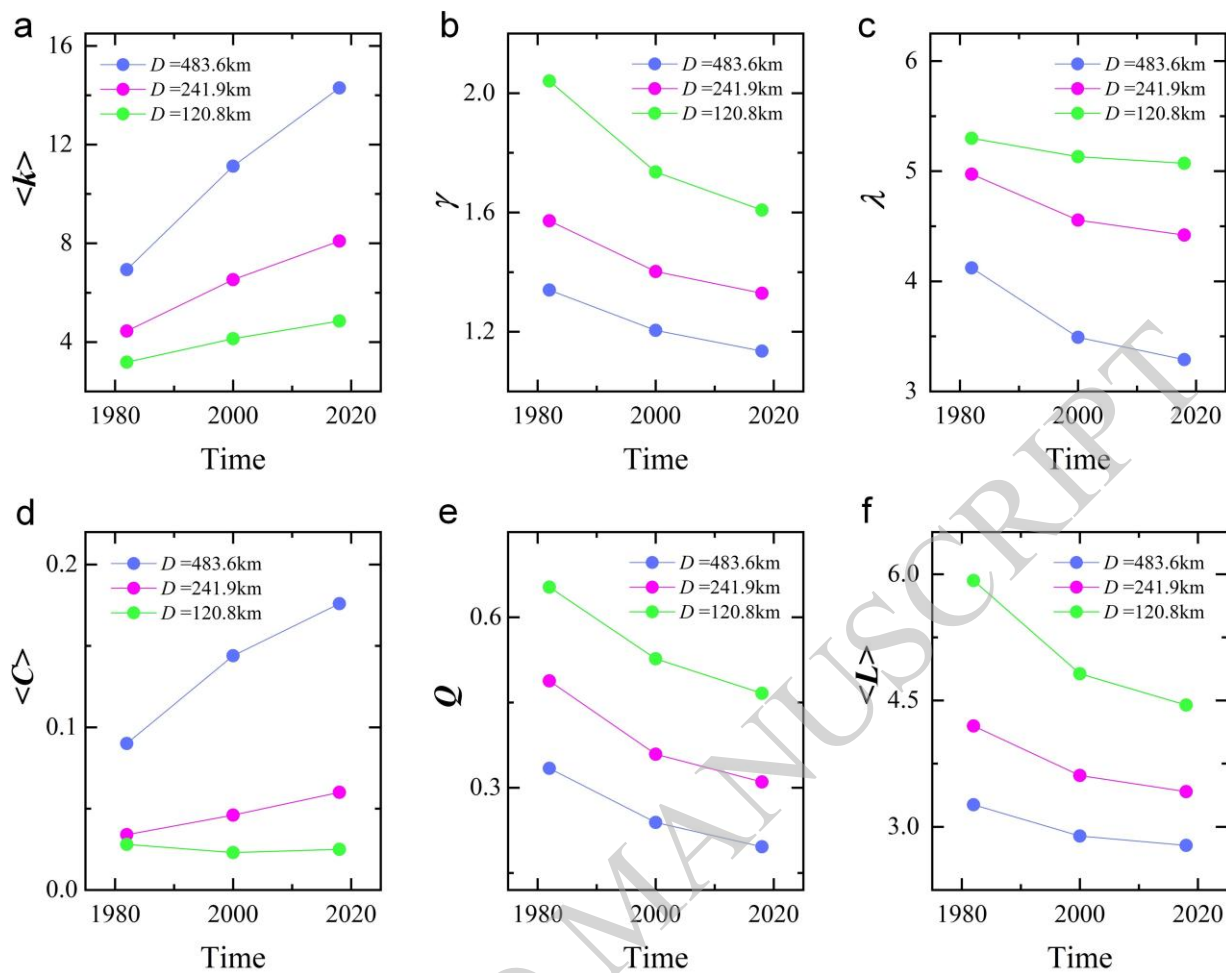


Table 1. Definition of symbols

Symbol	Definition
N	Number of events in the seismic dataset
R	Earth radius
D	Spatial scale of the spherical triangulation
d	Tessellation level of the spherical triangulation
V	Number of vertices in the earthquake network
E	Number of links in the earthquake network
v	Weight of a vertex in the earthquake network
w	Weight of a link in the earthquake network
$P(v)$	Distribution of vertex weights
$P(w)$	Distribution of link weights
θ	Collection of non-empty cells (containing at least one event) in the seismic dataset
$\langle k \rangle$	Average connectivity of the network
γ	Exponent of the power law relation between $P(v)$ and v
λ	Exponent of the power law relation between $P(w)$ and w
$\langle C \rangle$	Average clustering coefficient of the network
Q	Modularity of the network
$\langle L \rangle$	Average path length of the network

Table 2. Topological measures of global spatio-temporal earthquake networks for different time periods.

Tessellation scale	Measure	Period 1964–1982	Period 1964–2000	Period 1964–2018
$D = 483.6$ km	$\langle k \rangle$	6.935	11.122	14.298
	γ	1.340 ± 0.056	1.205 ± 0.035	1.135 ± 0.030
	λ	4.121 ± 0.105	3.491 ± 0.053	3.289 ± 0.058
	$\langle C \rangle$	0.090	0.144	0.176
	Q	0.334	0.239	0.196
	$\langle L \rangle$	3.262	2.891	2.779
$D = 241.9$ km	$\langle k \rangle$	4.449	6.526	8.089
	γ	1.572 ± 0.103	1.402 ± 0.049	1.329 ± 0.038
	λ	4.973 ± 0.384	4.556 ± 0.106	4.420 ± 0.048
	$\langle C \rangle$	0.034	0.046	0.060
	Q	0.488	0.359	0.310
	$\langle L \rangle$	4.198	3.610	3.417
$D = 120.8$ km	$\langle k \rangle$	3.188	4.133	4.856
	γ	2.041 ± 0.079	1.736 ± 0.054	1.608 ± 0.058
	λ	5.298 ± 0.300	5.131 ± 0.145	5.071 ± 0.115
	$\langle C \rangle$	0.030	0.023	0.025
	Q	0.653	0.527	0.466
	$\langle L \rangle$	5.924	4.818	4.445

Table 3. Topological measures of earthquake networks for datasets with similar number of events.

Tessellation scale	Measure	Events 1–2578	Events 2579–5157	Events 5158–7736
$D = 483.6$ km	Nodes V	567	539	551
	Edges E	2252	2257	2223
	$\langle k \rangle$	7.570	7.922	7.641
	γ	1.525±0.088	1.523±0.080	1.509±0.104
	λ	3.703±0.151	3.875±0.095	3.653±0.221
	$\langle C \rangle$	0.087	0.096	0.094
	Q	0.319	0.307	0.322
	$\langle L \rangle$	3.184	3.076	3.132
$D = 241.9$ km	Nodes V	958	909	951
	Edges E	2378	2385	2370
	$\langle k \rangle$	4.720	4.953	4.707
	γ	1.901±0.139	1.986±0.119	2.033±0.093
	λ	3.587±0.277	3.807±0.259	3.450±0.252
	$\langle C \rangle$	0.036	0.026	0.032
	Q	0.471	0.445	0.475
	$\langle L \rangle$	4.083	3.938	4.091
$D = 120.8$ km	Nodes V	1431	1389	1434
	Edges E	2474	2479	2462
	$\langle k \rangle$	3.296	3.374	3.259
	γ	2.459±0.135	2.580±0.122	2.607±0.151
	λ	4.590±0.369	4.807±0.254	4.388±0.285
	$\langle C \rangle$	0.029	0.015	0.020
	Q	0.633	0.614	0.646
	$\langle L \rangle$	5.716	5.493	5.815